

Table 1: Evaluation suites and protocol. Training steps and seeds are reported per configuration; the accompanying protocol defines the compared methods and evaluation modes.

Suite	Tasks	Budget(s)	Steps	Seeds	Compared methods
SafePO navigation	Point/Car: Goal, Button, Push (level 1); Goal, Button (level 2)	—	—	—	—
SafePO velocity	Ant, HalfCheetah, Hopper, Walker2d, Swimmer, Humanoid	—	—	—	—
Bullet Safety Gym	Ball/Car/Drone: Run and Circle	—	—	—	—
Hard + soft	CarGoal2, PointGoal2, RacecarGoal2	—	—	—	—
Controlled CMDPs	Exact tabular cells; neural-policy bridge; local-capacity grids	—	—	—	—

Table 2: Reward and hard/soft costs from the same evaluated policy. Each task cell reports return R and its 95% interval on the first line, then native contact percentage p with its reported 95% interval or upper bound and mean hazard cost H on the second. The soft budget is $H \leq 25$.

Method / execution	CarGoal2	PointGoal2	RacecarGoal2
SDAC	$R : \text{—} [\text{lo}, \text{hi}]$	$R : \text{—} [\text{lo}, \text{hi}]$	$R : \text{—} [\text{lo}, \text{hi}]$
native	$p : \text{—} [\text{lo}, \text{hi}]; H : \text{—}$	$p : \text{—} [\text{lo}, \text{hi}]; H : \text{—}$	$p : \text{—} [\text{lo}, \text{hi}]; H : \text{—}$
SDAC	$R : \text{—} [\text{lo}, \text{hi}]$	$R : \text{—} [\text{lo}, \text{hi}]$	$R : \text{—} [\text{lo}, \text{hi}]$
margin d	$p : \text{—} [\text{lo}, \text{hi}]; H : \text{—}$	$p : \text{—} [\text{lo}, \text{hi}]; H : \text{—}$	$p : \text{—} [\text{lo}, \text{hi}]; H : \text{—}$
SDAC	$R : \text{—} [\text{lo}, \text{hi}]$	$R : \text{—} [\text{lo}, \text{hi}]$	$R : \text{—} [\text{lo}, \text{hi}]$
budget $b = 0.5$	$p : \text{—} [\text{lo}, \text{hi}]; H : \text{—}$	$p : \text{—} [\text{lo}, \text{hi}]; H : \text{—}$	$p : \text{—} [\text{lo}, \text{hi}]; H : \text{—}$
SDAC	$R : \text{—} [\text{lo}, \text{hi}]$	$R : \text{—} [\text{lo}, \text{hi}]$	$R : \text{—} [\text{lo}, \text{hi}]$
shielded	$p : \text{—} [\text{lo}, \text{hi}]; H : \text{—}$	$p : \text{—} [\text{lo}, \text{hi}]; H : \text{—}$	$p : \text{—} [\text{lo}, \text{hi}]; H : \text{—}$
SDAC	$R : \text{—} [\text{lo}, \text{hi}]$	$R : \text{—} [\text{lo}, \text{hi}]$	$R : \text{—} [\text{lo}, \text{hi}]$
native	$p : \text{—} [\text{lo}, \text{hi}]; H : \text{—}$	$p : \text{—} [\text{lo}, \text{hi}]; H : \text{—}$	$p : \text{—} [\text{lo}, \text{hi}]; H : \text{—}$
SDAC	$R : \text{—} [\text{lo}, \text{hi}]$	$R : \text{—} [\text{lo}, \text{hi}]$	$R : \text{—} [\text{lo}, \text{hi}]$
margin d	$p : \text{—} [\text{lo}, \text{hi}]; H : \text{—}$	$p : \text{—} [\text{lo}, \text{hi}]; H : \text{—}$	$p : \text{—} [\text{lo}, \text{hi}]; H : \text{—}$
SDAC	$R : \text{—} [\text{lo}, \text{hi}]$	$R : \text{—} [\text{lo}, \text{hi}]$	$R : \text{—} [\text{lo}, \text{hi}]$
budget $b = 0.5$	$p : \text{—} [\text{lo}, \text{hi}]; H : \text{—}$	$p : \text{—} [\text{lo}, \text{hi}]; H : \text{—}$	$p : \text{—} [\text{lo}, \text{hi}]; H : \text{—}$
SDAC	$R : \text{—} [\text{lo}, \text{hi}]$	$R : \text{—} [\text{lo}, \text{hi}]$	$R : \text{—} [\text{lo}, \text{hi}]$
shielded	$p : \text{—} [\text{lo}, \text{hi}]; H : \text{—}$	$p : \text{—} [\text{lo}, \text{hi}]; H : \text{—}$	$p : \text{—} [\text{lo}, \text{hi}]; H : \text{—}$
Saute	$R : \text{—} [\text{lo}, \text{hi}]$	$R : \text{—} [\text{lo}, \text{hi}]$	$R : \text{—} [\text{lo}, \text{hi}]$
unfiltered	$p : \text{—} [\text{lo}, \text{hi}]; H : \text{—}$	$p : \text{—} [\text{lo}, \text{hi}]; H : \text{—}$	$p : \text{—} [\text{lo}, \text{hi}]; H : \text{—}$
ET-MDP	$R : \text{—} [\text{lo}, \text{hi}]$	$R : \text{—} [\text{lo}, \text{hi}]$	$R : \text{—} [\text{lo}, \text{hi}]$
unfiltered	$p : \text{—} [\text{lo}, \text{hi}]; H : \text{—}$	$p : \text{—} [\text{lo}, \text{hi}]; H : \text{—}$	$p : \text{—} [\text{lo}, \text{hi}]; H : \text{—}$
Safety Layer	$R : \text{—} [\text{lo}, \text{hi}]$	$R : \text{—} [\text{lo}, \text{hi}]$	$R : \text{—} [\text{lo}, \text{hi}]$
projected actions	$p : \text{—} [\text{lo}, \text{hi}]; H : \text{—}$	$p : \text{—} [\text{lo}, \text{hi}]; H : \text{—}$	$p : \text{—} [\text{lo}, \text{hi}]; H : \text{—}$
FDPI	$R : \text{—} [\text{lo}, \text{hi}]$	$R : \text{—} [\text{lo}, \text{hi}]$	$R : \text{—} [\text{lo}, \text{hi}]$
untyped	$p : \text{—} [\text{lo}, \text{hi}]; H : \text{—}$	$p : \text{—} [\text{lo}, \text{hi}]; H : \text{—}$	$p : \text{—} [\text{lo}, \text{hi}]; H : \text{—}$
Typed (ours)	$R : \text{—} [\text{lo}, \text{hi}]$	$R : \text{—} [\text{lo}, \text{hi}]$	$R : \text{—} [\text{lo}, \text{hi}]$
unfiltered	$p : \text{—} [\text{lo}, \text{hi}]; H : \text{—}$	$p : \text{—} [\text{lo}, \text{hi}]; H : \text{—}$	$p : \text{—} [\text{lo}, \text{hi}]; H : \text{—}$
Typed (ours)	$R : \text{—} [\text{lo}, \text{hi}]$	$R : \text{—} [\text{lo}, \text{hi}]$	$R : \text{—} [\text{lo}, \text{hi}]$
candidate selector	$p : \text{—} [\text{lo}, \text{hi}]; H : \text{—}$	$p : \text{—} [\text{lo}, \text{hi}]; H : \text{—}$	$p : \text{—} [\text{lo}, \text{hi}]; H : \text{—}$

TABLE LAYOUT PROOF

This page checks the template only. Dashes and interval labels are placeholders, not empirical results. Production builds reject missing values.